DATABRICKS · UPSTREAM SUBSURFACE

Subsurface Intelligence

From raw well logs to a located, valued, governed drilling decision, on one platform, in one sitting.

Locate → Read → Fill the sonic → Model in voxels → Value → Decide

Working demo on Databricks · Lakeflow · Unity Catalog · MLflow · Model Serving · AI Gateway · Spatial SQL · Genie · Apps

01 · THE DECISION

Where does the next well go, and is it worth drilling? The answer is spread across five systems.

DATA IN SILOS

Logs, tops, OSDU records, economics, maps

Each lives with a different team and tool. Every new question starts with an export and a hand-off.

GAPS IN THE RECORD

Wells drilled without a key curve

A missing sonic log blocks geomechanics and seismic ties. The usual fix is an empirical relation applied by hand.

AI WITHOUT GUARDRAILS

Assistants that cannot see, or ignore, entitlements

Either the model has no access to the governed data, or it has too much and partners see what they should not.

Subsurface Intelligence puts the logs, the geomodel, the economics and the AI on one governed platform, so the decision is one workflow instead of five projects.

Six steps, one platform, every number traceable to a governed table

#	STEP	WHAT HAPPENS	DATABRICKS
1	Locate the wells	Fleet map with basin footprints and well spacing; the voxel model's proposed step-out per field	Spatial SQL (ST_ConvexHull, ST_Distance), H3
2	Read the logs	Raw curves, QC flags, corrections, Vcl / porosity / Archie Sw	Lakeflow SDP bronze → silver → gold
3	Fill the missing sonic	A served model predicts DT where none was logged, scored on held-out wells	MLflow, UC model registry, Model Serving, AI Gateway
4	Model the subsurface	49,152-cell voxel geomodel kriged from the logs, with uncertainty, ranks a step-out location	Apps, Unity Catalog
5	Value the well	NPV10, IRR, break-even against WTI	Certified UC functions, metric views
6	Decide with AI	Supervisor plans specialists, cites evidence, meters cost; Genie answers in plain English	Agents, AI Gateway, Genie

The model has to beat the method engineers already trust, on wells it never saw

METHOD · LEAVE-ONE-WELL-OUT	MAE MS/FT	RMSE	R^2
Gardner, published coefficients	17.23	22.25	-1.54
Gardner, refit on training wells	10.68	12.34	0.22
Faust, refit on training wells	6.64	8.66	0.62
Gradient-boosted trees, all curves	3.05	3.81	0.925
Linear, all curves · served	**3.00**	**3.76**	**0.927**

13 wells, 31,189 samples. Whole wells are held out because samples from one well share its geology; a random split would leak. Champion rule: lowest leave-one-well-out MAE; within 2% the simpler model wins.

3.6x

lower error than Gardner calibrated on the same wells

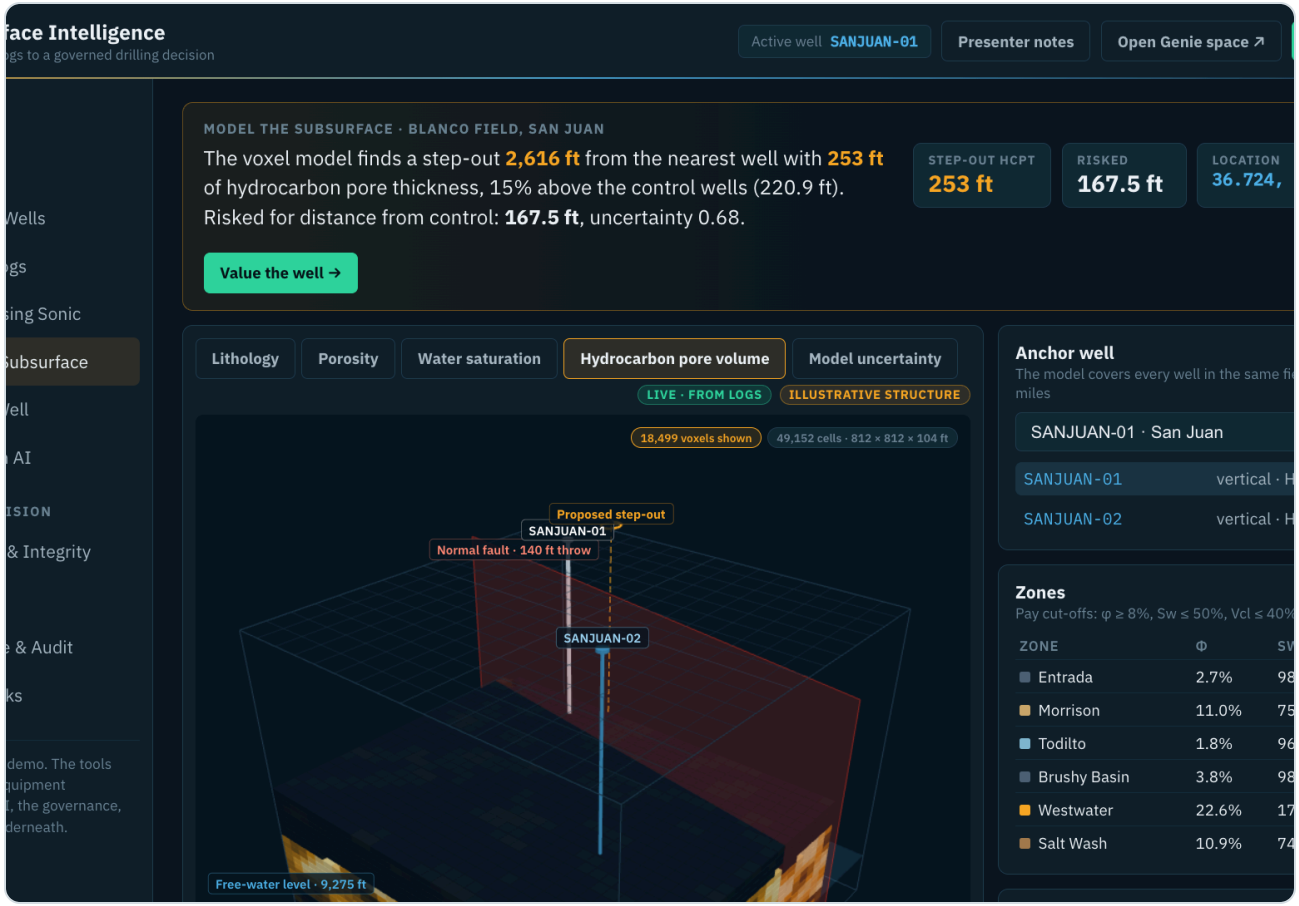
3.0

µs/ft mean error on unseen wells

DECISION WORTH DEFENDING

Trees and linear tie at the noise floor, so the simpler model is served: easier to explain to a petrophysicist, cheaper to run, nothing lost. It lives in Unity Catalog as **synthetic_sonic@champion** behind AI Gateway, and the app falls back to Gardner, labelled, if the endpoint is cold.

A voxel geomodel from the logs in under a second, honest about what it does not know



253 ft

hydrocarbon pore thickness at the step-out, 15% above the control wells (221 ft)

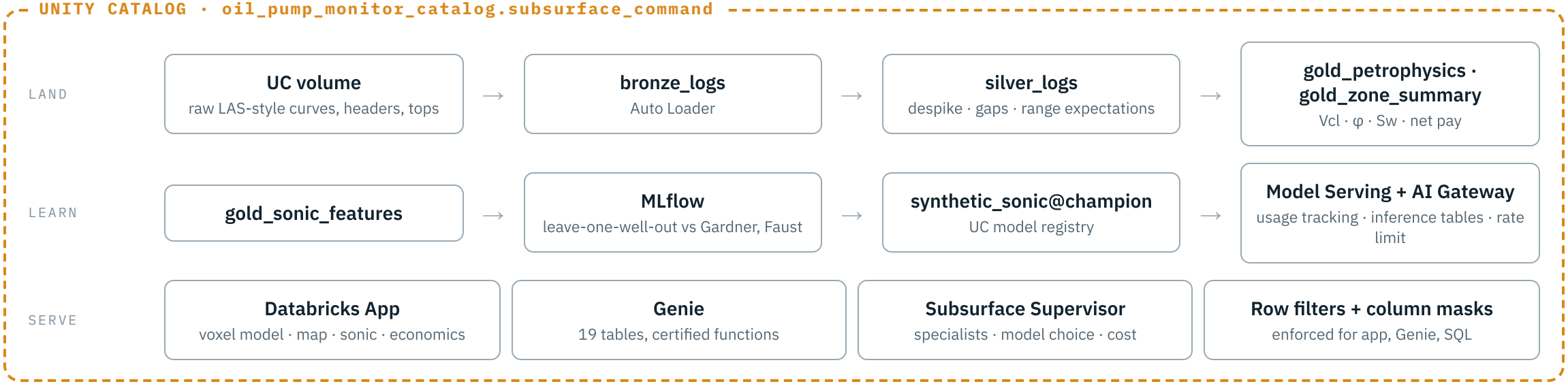
168 ft

risked for distance from control (uncertainty 0.68)

- Structure honours every well top; dip, closure and one normal fault are a modeled framework `Illustrative`
- Vcl, porosity and Sw from the logs, kriged between wells per stratigraphic sub-layer `Live`
- Reservoir Sw follows a saturation-height function above the free-water level, so the crest carries more oil than the flank
- Step-out 2,616 ft from SANJUAN-01, ranked on kriged HCPT and haircut by uncertainty

05 · HOW IT IS BUILT

One pipeline, one registry, one governance layer



One Asset Bundle job, **subsurface-command-build**, lands the data, runs the pipeline, trains, registers, serves and scores. Rebuilding the whole stack is one command.

06 · GOVERN THE DATA AND THE AI

The same policy holds in the app, in Genie, and in the SQL editor, because it lives in Unity Catalog

ROW FILTER

JV partners see only their basin

gold_zone_summary: an operator sees 84 rows across 4 basins; mapped as a San Juan JV partner, the same query returns 30 rows, San Juan only.

COLUMN MASK

Interpretation stays in-house

Sw and hydrocarbon pore feet return NULL to the partner while net pay stays visible. One function, applied to every consumer.

AI CONTROL

Choice, cost, control

Swap models through AI Gateway with no code change, see token cost per run, guardrails on every call, and agents running as the signed-in user.

Governance is not an app feature you have to trust; it is the platform. Ask Genie the same question as the partner and the masked columns stay masked.

Every screen says where its numbers come from

ELEMENT	STATUS	DETAIL
Pipeline, model, endpoint, policies, Genie	Live	Running on Databricks; job, endpoint and policies verified
Sonic predictions	Live	Served model on every request; labelled Gardner fallback if cold
Voxel properties and step-out	Live	Computed in the app from the wells' logs and tops on every load
Well logs	Synthetic	Physically coupled rock model (porosity, shale, fluids); your logs replace them in the pilot
Structural framework	Illustrative	Dip, closure and fault are modeled, honoured at the tops; a real horizon model plugs in
OSDU analog panels	Offline	Fed by a partner ADME workspace that is not live; the panels say so

Prove it on one field with your logs, then reuse the template

WEEKS 1–2

Land

Your LAS files, tops and headers for one field into a UC volume; same pipeline, your data.

WEEKS 3–4

Score

Retrain the sonic model on your wells; agree the bar it must clear against your current method.

WEEKS 5–8

Decide

Voxel model on your horizons; one step-out reviewed by your geoscientists against their own pick.

AFTER

Scale

Next field, next missing curve, partner access under the same policies.

ASK 1 · BUSINESS

Name one field and the person who owns its next-well decision

Success is a step-out your team agrees is credible, with its value, in weeks instead of a quarter.

ASK 2 · TECHNICAL

A two-week working session with petrophysics and data

Map your logs, tops and OSDU records to the pipeline, and agree the acceptance bar for the sonic model.

APPENDIX · THE APP

What you will see

